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LEARNER GUIDE

For

Mastering Semiconductor Key Modules for Enhanced CMOS Process Integration and Device Performance

TGS Ref No: TGS-2024051248

Conducted by

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DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
8.0	20 August 2026	New Learner Guide built from the complete v7 trainer-deck coverage; aligned to eight activities, current WSQ administration and v8.0 publication.	Dr Alfred Ang

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How to Use This Guide

This guide holds the detailed procedures intentionally omitted from the concept-led slide deck. Work through the eight activities in sequence, keep evidence in the named folder, and use acceptance criteria before moving to the next integration decision.

Field	Details
Course	Mastering Semiconductor Key Modules for Enhanced CMOS Process Integration and Device Performance
Course code	TGS-2024051248
TSC	Process Integration · ELE-SIS-4002-1.1
Duration	16 hours over 2 days
Assessment	Written Assessment (WA-SAQ) - 2 open-ended questions, K1-K2, 1 hour, open book. Practical Performance (PP) - 2 open-ended tasks, A1-A5, 1 hour, open book.
LMS	https://lms-tms.tertiaryinfotech.com/

Learning and Assessment Alignment

- LO1: Identify process flows and modules for specific CMOS technologies in manufacturing.
- LO2: Determine CMOS device performance and verify process integration in accordance with product specifications.

Knowledge and Ability Criteria

- K1: Specific technology process flow
- K2: Types of performance metrics
- A1: Identify process flow for the specific technology in the manufacturing process in accordance with technology specifications
- A2: Determine process interaction between and within process modules to the specific technology in accordance with technology specifications
- A3: Determine device performance in accordance with product specifications
- A4: Verify process integration performance to the specific technology in accordance with product specifications
- A5: Determine follow-up actions required

Before You Start - What You Need

- The current v8.0 learner slides and this Learner Guide.
- The complete activities/ folder with Markdown/PDF resources and Excel workbooks.
- Microsoft Excel or compatible spreadsheet software with automatic calculation enabled.

- A drawing tool or A3 paper for process maps and integration diagrams.
- Access to the LMS for courseware download and assessment submission.

Topic 1 - CMOS Process Modules and Process Flow

Wafer fab and cleanroom control

A wafer fab is a controlled manufacturing system in which cleanroom, facility, tool, material and operator controls protect very small device features.

Contamination can affect device-processing yield, performance and reliability. Particle, ionic, molecular, water, chemical and equipment sources therefore require separate controls.

Module ownership does not remove integration ownership: incoming surface condition, outgoing measurable characteristics and hold/release logic must be explicit.

Photolithography and masks

Photolithography transfers a mask/reticle pattern into photoresist using controlled exposure, focus, bake and develop conditions.

Positive and negative resists respond differently; mask field polarity and resist tone decide whether the wafer receives a hole or island pattern.

Resolution, critical dimension, aspect ratio, resist sensitivity, depth of focus, alignment and defectivity interact. Shorter wavelength and higher numerical aperture improve resolution but can reduce process margin.

OPC, phase-shift masks and antireflective coatings improve image fidelity; CD-SEM and overlay evidence verify the result.

Etch, strip and clean

Wet etch is commonly isotropic and can undercut the resist; dry plasma/RIE processes support anisotropic profiles and smaller features.

Selectivity protects underlying films. Over-etch can ensure clearing but can also damage stop layers or change feature profile.

Incomplete etch, undercut, microloading, residue and resist-strip damage require different evidence and corrective levers.

Ion implantation, diffusion and RTP

Implant dose and energy set concentration and depth; beam scanning, charging, channeling and shadowing affect uniformity and accuracy.

Implantation damages the crystal. Annealing repairs damage and activates dopant, while thermal budget also changes lateral/vertical diffusion and junction depth.

RTP provides fast ramp and short dwell for shallow junctions, while furnace cycles provide different uniformity and throughput trade-offs.

Film deposition and metallization

CVD, PECVD, epitaxy and PVD/sputtering supply different conformality, temperature, purity and material options.

Barrier/liner and seed continuity are critical in copper metallization; dual-damascene integration connects lithography, etch, deposition, electrofill and CMP.

Film thickness, composition, stress, conformality and interface condition must be verified before downstream release.

Chemical-mechanical polishing

CMP combines mechanical abrasion with slurry chemistry to create global planarity across multi-level structures.

Removal rate, selectivity, pad/slurry condition and within-wafer uniformity influence incomplete clear, dishing, erosion and recess.

Post-CMP cleaning protects the next module from abrasive particles, metal contamination and residue.

Integrated CMOS process flow

Isolation and well formation establish device regions; gate and source/drain modules form transistors; contacts and interconnect create the BEOL network.

Each mask level is governed by design rules, alignment, CD and defect requirements. The integration flow must include inspection, metrology and feedback gates.

A process flow is not complete until it names evidence, limits, ownership and the action taken when a result is not acceptable.

Topic 2 - CMOS Process Integration and Performance

Yield, metrology and electrical performance

Wafer-fabrication yield, wafer-sort yield and packaging yield multiply into overall yield. Defect density, die area, process variation, mask defects and breakage can limit different stages.

Ellipsometry, stylus profilometry, SEM, TEM, CD measurement, AFM and SIMS answer different thickness, topography, profile, composition and concentration questions.

Four-point probe, gate-oxide tests and device I-V measurements connect physical process results to product specifications.

Reliability performance

Hot-carrier effects, time-dependent dielectric breakdown and electromigration are key reliability mechanisms retained from the legacy course.

Reliability evidence combines stress conditions, time-dependent response, failure criteria and physical confirmation; one electrical pass does not prove lifetime performance.

SPC and follow-up action

Control limits describe stable process behaviour; specification limits describe product requirements. They answer different questions.

Cp describes potential capability from spread; Cpk also considers centering. Capability is meaningful only after stability is established.

Xbar-R/Xbar-S charts monitor variables; p/np/c/u charts monitor attribute counts or proportions under the appropriate sample-size conditions.

An out-of-control response proceeds through containment, stratification, confirmation, correction, verification and documented follow-up.

Packaging and future technologies

Packaging includes wafer thinning, die separation, die attach, wire/flip-chip bonding, sealing, plating, final test and burn-in.

Heterogeneous integration and chiplets combine dies using advanced interconnect and packaging evidence; known-good-die, thermal and mechanical concerns become integration inputs.

FinFET, GAAFET, vertical devices and 2D materials change geometry and material requirements, but still require process capability, metrology, device, reliability and yield gates.

Detailed Step-by-Step Activities

Activity 1 - Build a CMOS Process Module Map

Field	Details
Goal	Organise the legacy module coverage into a defensible FEOL-to-BEOL flow and expose the interfaces that can propagate variation.
Scenario	A new 300 mm logic product is moving from process development into a pilot manufacturing line. The integration team needs one agreed process map before module owners set recipes and control plans.
Alignment	LO1 · K1, A1, A2
Duration	45 minutes
Folder	activities/activity-01-process-module-map/

What You Need

- Activity worksheet and evidence template in this folder
- Learner slides covering wafer fab, photolithography, etch, implant, thermal, film, CMP and integrated flows
- A3 paper or a digital drawing tool

Step-by-Step

1. Open the worksheet and write the target product assumptions: substrate type, logic technology family and the intended final device outcome.
2. Create seven module lanes: photolithography, etch/clean, ion implantation, diffusion/RTP, film deposition, CMP and metrology/control.
3. Place the major FEOL stages in order: wafer preparation, isolation, well formation, gate stack, source/drain formation and contacts.
4. Add the BEOL stages: dielectric deposition, via/trench patterning, barrier/seed, metal fill, CMP, passivation and wafer sort.
5. For each hand-off, identify the incoming measurable characteristic, such as CD, overlay, film thickness, dose, sheet resistance or surface planarity.
6. Mark at least three feedback loops where metrology can cause hold, rework, recipe correction or engineering disposition.
7. Circle two interactions where a change in one module can create a downstream electrical or yield effect.
8. Check the map against the provided process-flow slides and correct missing or duplicated stages.
9. Export or photograph the final map and name the evidence file A01-process-map.

Acceptance Criteria

- [ ] The map contains all seven key process modules and distinguishes FEOL from BEOL.
- [ ] At least eight process hand-offs have a measurable characteristic.
- [ ] At least three control/feedback loops and two cross-module interactions are justified.

Reflection and Evidence

- Which module has the greatest ability to propagate a hidden defect downstream, and why?
- What evidence would allow the integration owner to release the lot to the next module?

Save the completed evidence with the A01- prefix and record the final release/hold/recommendation decision in evidence-template.pdf or evidence-template.md.

Activity 2 - Diagnose a Pattern-Transfer Process Window

Field	Details
Goal	Connect mask/resist/exposure decisions to etch profile and select a stable corrective action without masking the root cause.
Scenario	A contact layer shows intermittent bridging and open defects. Lithography CD, resist profile and plasma-etch selectivity must be reviewed together rather than as isolated module results.
Alignment	LO1 · K1, A2
Duration	60 minutes

Field	Details
Folder	activities/activity-02-pattern-transfer-window/

What You Need

- Pattern-transfer worksheet and process-window workbook
- Legacy diagrams on resist polarity, Rayleigh resolution, OPC, selectivity and anisotropy
- Calculator or spreadsheet software

Step-by-Step

1. Review the defect images in the worksheet and classify each as bridge, open, incomplete etch, undercut or resist-collapse risk.
2. Open pattern-transfer-window.xlsx and read the focus, exposure dose, post-develop CD and post-etch CD observations.
3. Mark cells that violate the target post-etch CD range of 88-112 nm.
4. Compare the lithography CD bias with the post-etch CD bias to decide whether the dominant shift occurs before or during etch.
5. Check whether edge cells fail more often than centre cells; record the spatial signature.
6. Select one lithography lever (focus, dose, resist thickness or bake) and one etch lever (gas ratio, bias power, pressure or over-etch).
7. Predict the intended effect and one possible side effect for each lever.
8. Write a confirmation plan using CD-SEM before etch, CD-SEM after etch and profile inspection on a cross-section.
9. Record a release decision: accept, conditional release, hold for engineering, or scrap.

Acceptance Criteria

- [ ] Every defect is linked to a plausible lithography/etch mechanism.
- [ ] The dominant CD-shift stage is supported by before/after evidence.
- [ ] Corrective levers include predicted benefit, risk and verification measurement.

Reflection and Evidence

- Why can improving resolution reduce depth of focus?
- When does additional over-etch protect yield, and when does it create reliability risk?

Save the completed evidence with the A02- prefix and record the final release/hold/recommendation decision in evidence-template.pdf or evidence-template.md.

Activity 3 - Balance Implantation and Thermal Budget

Field	Details
Goal	Create an implant-to-anneal decision trail that balances activation, junction depth, damage recovery and channeling risk.
Scenario	After a source/drain extension implant, sheet

Field	Details
	resistance is above target and short-channel control is drifting. The team must distinguish dose/energy error from activation and diffusion effects.
Alignment	LO1 · K1, A2, A3
Duration	60 minutes
Folder	activities/activity-03-implant-thermal-budget/

What You Need

- Implant and RTP data cards in the worksheet
- Legacy diagrams for beam scanning, channeling, damage and RTP
- Evidence template

Step-by-Step

1. Read the target dose, implant energy, tilt/twist, sheet resistance and junction-depth specifications.
2. Compare actual dose and energy with the target and note whether the deviation can explain the electrical signature.
3. Inspect the channeling indicator and decide whether tilt/twist or an amorphising layer should be reviewed.
4. Use the damage/activation card to separate as-implanted lattice damage from electrically active dopant.
5. Compare the furnace and RTP thermal cycles by peak temperature, dwell time and ramp rate.
6. Predict how each thermal cycle affects activation, lateral diffusion and junction depth.
7. Choose a containment action for the current lot and a recipe action for the next engineering split.
8. Define the minimum verification set: sheet resistance map, SIMS profile or junction-depth proxy, and electrical monitor.
9. Complete the risk statement: action, expected benefit, possible adverse effect and stop condition.

Acceptance Criteria

- [ ] The recommendation distinguishes implant dose/energy, damage and activation mechanisms.
- [ ] Thermal-budget effects on activation and diffusion are explicit.
- [ ] Follow-up measurements can confirm or reject the proposed cause.

Reflection and Evidence

- Why does a shorter high-temperature RTP cycle often protect shallow junctions?
- How can a sheet-resistance improvement still hide a device-performance problem?

Save the completed evidence with the A03- prefix and record the final release/hold/recommendation decision in evidence-template.pdf or evidence-template.md.

Activity 4 - Integrate Film Deposition and CMP

Field	Details
Goal	Trace deposition, fill and CMP interactions and design a verification plan that protects both planarity and electrical performance.
Scenario	A dual-damascene copper level meets line-resistance targets at wafer centre but fails near the edge with dishing, erosion and suspected barrier continuity loss.
Alignment	LO1 · K1, A2, A4
Duration	60 minutes
Folder	activities/activity-04-film-cmp-integration/

What You Need

- Cross-section cards and film/CMP worksheet
- Legacy diagrams for barrier/seed, copper fill, slurry, polishing and post-CMP clean
- Evidence template

Step-by-Step

1. Label the dielectric, barrier/liner, seed, copper fill and overburden layers on the supplied cross-section.
2. Identify which layer prevents diffusion and which layer enables electroplating continuity.
3. Compare centre and edge thickness/planarity observations and classify dishing, erosion or incomplete clear.
4. Trace how PVD coverage or electroplating voids can change the apparent CMP result.
5. Choose two CMP inputs to review: downforce, platen speed, pad condition, slurry chemistry, flow or polish time.
6. For each input, state the expected effect on removal rate, selectivity and within-wafer uniformity.
7. Add a post-CMP clean and contamination-control check to the integration plan.
8. Define metrology at three points: pre-CMP overburden, post-CMP topography and electrical line/via continuity.
9. Write a hold/release rule based on planarity plus electrical evidence.

Acceptance Criteria

- [ ] The layer stack and each module's function are correctly labelled.
- [ ] The analysis distinguishes deposition/fill causes from polishing causes.
- [ ] The release rule uses both physical and electrical evidence.

Reflection and Evidence

- Why can a visually planar surface still fail electrical verification?
- Which measurement best distinguishes dishing from incomplete copper clear?

Save the completed evidence with the A04- prefix and record the final release/hold/recommendation decision in evidence-template.pdf or evidence-template.md.

Activity 5 - Review an STI Integration Flow

Field	Details
Goal	Produce the process-flow and challenge analysis used as the foundation for Practical Performance Task 1.
Scenario	TechSemicon is introducing shallow trench isolation for an advanced CMOS product. Electrical spread and active-area CD variation appear after integration, and the review board needs a complete process/evidence chain.
Alignment	LO2 · A1, A2, A4
Duration	75 minutes
Folder	activities/activity-05-sti-integration-review/

What You Need

- STI process cards, worksheet and evidence template
- Legacy LOCOS/STI, lithography, etch, CVD and CMP slides
- A3 paper or digital flow-diagram tool

Step-by-Step

1. Arrange the STI cards in order: pad oxide, nitride deposition, lithography, trench etch, liner oxidation, trench fill, CMP and nitride strip.
2. For each step, write the purpose, key input and outgoing measurable characteristic.
3. Add pre-clean and post-CMP clean controls at the correct points.
4. Mark the interfaces where active-area CD, trench depth/profile, voiding, dishing/erosion and stress can be introduced.
5. Connect each risk to one suitable measurement such as CD-SEM, cross-section SEM/TEM, film thickness, topography or electrical isolation test.
6. Compare STI with LOCOS and state why STI supports scaling despite its integration challenges.
7. Choose two control points for a pilot-line hold/release plan.
8. Draw the final simplified flow without referring to the cards.
9. Use the checklist to review completeness, sequence and evidence traceability.

Acceptance Criteria

- [ ] The eight core STI stages are in the correct order and have a purpose.
- [ ] At least four implementation challenges are linked to measurements.
- [ ] The flow explains both scaling benefit and integration risk.

Reflection and Evidence

- Which STI step most directly controls active-area CD?
- How can CMP-induced topography alter a later lithography process window?

Save the completed evidence with the A05- prefix and record the final release/hold/recommendation decision in evidence-template.pdf or evidence-template.md.

Activity 6 - Analyse Yield and Process Capability

Field	Details
Goal	Recreate and improve the legacy ProcessCapability exercise using an auditable Excel workbook with dynamic formulas.
Scenario	A gate module has a target CD of 100 nm with specification limits at 90 and 110 nm. The process owner needs Cp/Cpk evidence and a reasoned yield-risk statement before qualification.
Alignment	LO2 · K2, A3, A4
Duration	75 minutes
Folder	activities/activity-06-yield-capability-analysis/

What You Need

- ProcessCapability.xlsx in this folder
- Legacy Cp/Cpk, sigma, yield and defect-density slides
- Worksheet and evidence template

Step-by-Step

1. Open ProcessCapability.xlsx and read the Instructions sheet before changing any inputs.
2. Go to Data and confirm the 30 hardcoded CD observations are shown in blue text.
3. Review the Analysis formulas for mean, sample standard deviation, Cp and Cpk; do not replace formulas with values.
4. Compare the mean with the 100 nm target and describe accuracy (centering).
5. Compare the standard deviation with the specification width and describe precision (spread).
6. Interpret Cp as potential capability and Cpk as actual capability considering centering.
7. Change one blue input value to 118 nm and observe how the metrics and out-of-spec count recalculate; then restore the value.
8. Use the histogram and capability summary to decide capable, conditionally capable or not capable.
9. Record one containment action and one long-term process action.
10. Export the Analysis sheet to PDF or capture a screenshot named A06-capability-evidence.

Acceptance Criteria

- [ ] Workbook formulas remain intact and show no Excel errors.
- [ ] The learner distinguishes Cp from Cpk and links both to process evidence.
- [ ] The decision includes containment, corrective action and verification.

Reflection and Evidence

- Why can Cp be acceptable while Cpk is unacceptable?
- Why does a capability index not prove that the process is statistically stable?

Save the completed evidence with the A06- prefix and record the final release/hold/recommendation decision in evidence-template.pdf or evidence-template.md.

Activity 7 - Respond to an SPC CD Excursion

Field	Details
Goal	Recreate the legacy P-chart/SPC exercises and generate the corrective-action chain used in Practical Performance Task 2.
Scenario	STI active-area CD is drifting across production lots. A point crosses the upper control limit and a seven-point upward run appears before the alarm. The team must contain risk and restore control.
Alignment	LO2 · K2, A3, A4, A5
Duration	90 minutes
Folder	activities/activity-07-spc-cd-excursion/

What You Need

- SPCdata.xlsx and p_Chart.xlsx in this folder
- Legacy control-limit, Xbar/R, p/c/u-chart and out-of-control rule slides
- Corrective-action worksheet and evidence template

Step-by-Step

1. Open SPCdata.xlsx and review the raw subgroup CD data, timestamps, tool IDs and lot IDs.
2. Check that subgroup means and ranges use formulas and that the chart updates when a blue input changes.
3. Identify the first out-of-control point and the earlier non-random run signal.
4. Separate control limits from product specification limits in the worksheet.
5. Write the immediate containment actions: stop/hold affected lots, preserve evidence, notify ownership and prevent automatic release.
6. Stratify the data by tool/chamber, wafer position, product and time to narrow the source.
7. Review lithography focus/dose, etch endpoint/profile and metrology calibration as candidate causes.

8. Define confirmation measurements using pre-etch CD, post-etch CD and cross-section profile evidence.
9. Select a corrective action and a controlled engineering split; state the rollback trigger.
10. Verify recovery using consecutive stable subgroups and an updated capability check.
11. Open p_Chart.xlsx and repeat the logic for variable sample sizes using defect proportion rather than CD measurements.
12. Export the SPC evidence and complete the follow-up action record.

Acceptance Criteria

- [ ] The response identifies both limit violation and non-random pattern.
- [ ] Containment precedes root-cause adjustment and protects affected lots.
- [ ] Corrective action, verification criteria and follow-up owner are explicit.

Reflection and Evidence

- Why must the team investigate a non-random run even when all points remain inside control limits?
- Which chart is appropriate for CD measurements, and which is appropriate for defect proportions?

Save the completed evidence with the A07- prefix and record the final release/hold/recommendation decision in evidence-template.pdf or evidence-template.md.

Activity 8 - Run a Future-Technology Integration Gate

Field	Details
Goal	Combine the entire course into a structured technology-gate recommendation without treating future devices as isolated marketing concepts.
Scenario	A product team is deciding whether to adopt chiplet packaging, FinFET or GAAFET for its next platform. Integration readiness must be assessed using process, performance, reliability and packaging evidence.
Alignment	LO2 · K1, K2, A2, A3, A4, A5
Duration	60 minutes
Folder	activities/activity-08-technology-integration-gate/

What You Need

- Technology-gate worksheet and evidence template
- Legacy packaging, heterogeneous integration, chiplet, FinFET, GAAFET, vertical transistor and 2D-material slides
- Outputs from Activities 1-7

Step-by-Step

1. Choose one candidate: chiplet integration, FinFET scaling or GAAFET scaling.
2. List the required process modules and identify which existing modules need a new material, geometry, tool or control method.
3. Map three critical cross-module interactions and the performance metric each can affect.
4. Define device metrics (for example leakage, drive current, resistance or capacitance) and reliability metrics relevant to the candidate.
5. Add packaging/interconnect implications, including thermal, mechanical and known-good-die considerations where applicable.
6. Create a five-row evidence gate: process capability, metrology readiness, electrical performance, reliability and yield/manufacturability.
7. Set a pass/conditional/fail threshold for every gate.
8. Identify the largest uncertainty and design the smallest engineering experiment that can reduce it.
9. Present a recommendation: proceed, proceed with conditions, or defer.
10. Record the next owner, due date and evidence required to close the decision.

Acceptance Criteria

- [ ] The recommendation integrates process, device, reliability, yield and packaging evidence.
- [ ] Every gate has a measurable threshold and an accountable follow-up action.
- [ ] The conclusion is conditional on evidence rather than technology hype.

Reflection and Evidence

- Which legacy process controls remain useful for the chosen technology?
- What new metrology or reliability evidence is essential before volume manufacturing?

Save the completed evidence with the A08- prefix and record the final release/hold/recommendation decision in evidence-template.pdf or evidence-template.md.

Quick Reference

Field	Details
Pattern transfer	Mask/resist -> expose/develop -> inspect CD -> etch -> strip/clean -> verify profile
Implant/thermal	Dose/energy/tilt -> implant -> damage/channeling check -> anneal/RTP -> sheet resistance/profile verification
Film/CMP	Surface prep -> barrier/seed or film -> fill/deposition -> thickness/profile -> CMP -> clean -> electrical verification
SPC response	Contain -> stratify -> confirm -> correct -> verify -> document follow-up

Field	Details
Capability	Establish stability first; then compare Cp and Cpk against the approved threshold
Evidence	Use the measurement that can distinguish competing causes, not merely the easiest available measurement

Assessment Flow

1. TRAQOM - scan the digital-attendance QR code on the LMS.
2. Complete Assessment Digital Attendance.
3. Complete the Written Assessment, then the Practical Performance assessment.
4. Upload completed answers on the LMS.
5. Sign the Assessment Summary Record.

Courseware and assessment portal: https://lms-tms.tertiaryinfotech.com/

Support

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